

General form of a DE of order n :

$$F(x, y, y', \dots, y^{(n)}) = 0 \rightarrow \textcircled{1}$$

Normal form

$$y^{(n)} = f(x, y, y', \dots, y^{(n-1)}) \rightarrow \textcircled{2}$$

Eq ① is linear if F is a linear function of $y, y', \dots, y^{(n)}$

$$\text{general form: } a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = g(x) \rightarrow \textcircled{3}$$

$$\text{Differential operator } L = a_n(x) \frac{d^n}{dx^n} + a_{n-1}(x) \frac{d^{n-1}}{dx^{n-1}} + \dots + a_0(x)$$

$$\textcircled{3} \text{ can be expressed as } L[y] = g(x)$$

$$L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2] \Rightarrow \text{if } y_1, y_2 \text{ are solutions of } \textcircled{3}, \text{ then}$$

$\alpha y_1 + \beta y_2$ may not be the solution

for $\textcircled{3}$. (RHS $\neq g(x)$)

if $y_1 = h(x)$ is a solution to $\textcircled{3}$, then is $y_2 = \alpha h(x)$ also a solution? - No

A linear DE $L[y] = g(x)$ is homogeneous if $g(x) \equiv 0$ over the Interval I .

if y_1 is a solution αy_1 is also a solution

if y_1, y_2 are solutions, then $\alpha y_1 + \beta y_2$ are solutions.

2nd order linear & homogeneous DE :-

general form: $a_2(x) y'' + a_1(x) y' + a_0(x) y = 0$

Normal form: $y'' + p(x) y' + q(x) y = 0$ if $a_2(x) \neq 0$ at any point in interval I .

① if y_1, y_2 are solutions of ①, then $\alpha y_1 + \beta y_2$ are also solutions on the interval I .

linear independence of n functions:- $y_1(x) \dots y_n(x)$

Example:- $y_1(x) = \sin x$ & $y_2(x) = \cos x$ (linearly independent).

if $\exists c_1, c_2$ such that $c_1 y_1(x) + c_2 y_2(x) \equiv 0$ over the interval I .
Not all 0

if $\exists \alpha: y_1(x) = \alpha y_2(x)$ ($\alpha \neq 0$)

for any n : $y_1, \dots, y_n$ are linearly dependent if $\exists c_1, \dots, c_n$ not all 0 such that
 $c_1 y_1 + \dots + c_n y_n \equiv 0$ over the interval I .

if $y_1, \dots, y_n$ are solutions of DE, then their linear dep can be determined using
Wronskian. W($y_1, \dots, y_n$).

$n=2$:- $w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \rightarrow \textcircled{1}$

Normal form: $y'' + P(x)y' + Q(x)y = 0.$ $\rightarrow \textcircled{2}$